CONNECTING ROD

DESIGNED USING AUTOCAD

Submitted By

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Abstract

This project focuses on the design and modeling of a connecting rod using AutoCAD. The connecting rod is a mechanical component that transmits motion and force between the piston and crankshaft in an engine. A 3D model was created using various AutoCAD solid modeling commands. Orthographic views were generated to represent the object in standard engineering drawing format. The project demonstrates the application of CAD tools in mechanical component design and visualization.

Introduction

A connecting rod is a mechanical component used in internal combustion engines to connect the piston to the crankshaft. It converts the reciprocating motion of the piston into rotational motion of the crankshaft.

Importance

- Transfers force from piston to crankshaft
- Essential component in engines
- Must withstand high compressive and tensile loads

Software Used

Software	Purpose
AutoCAD 2026	2D Drafting and 3D Modeling

3D Model

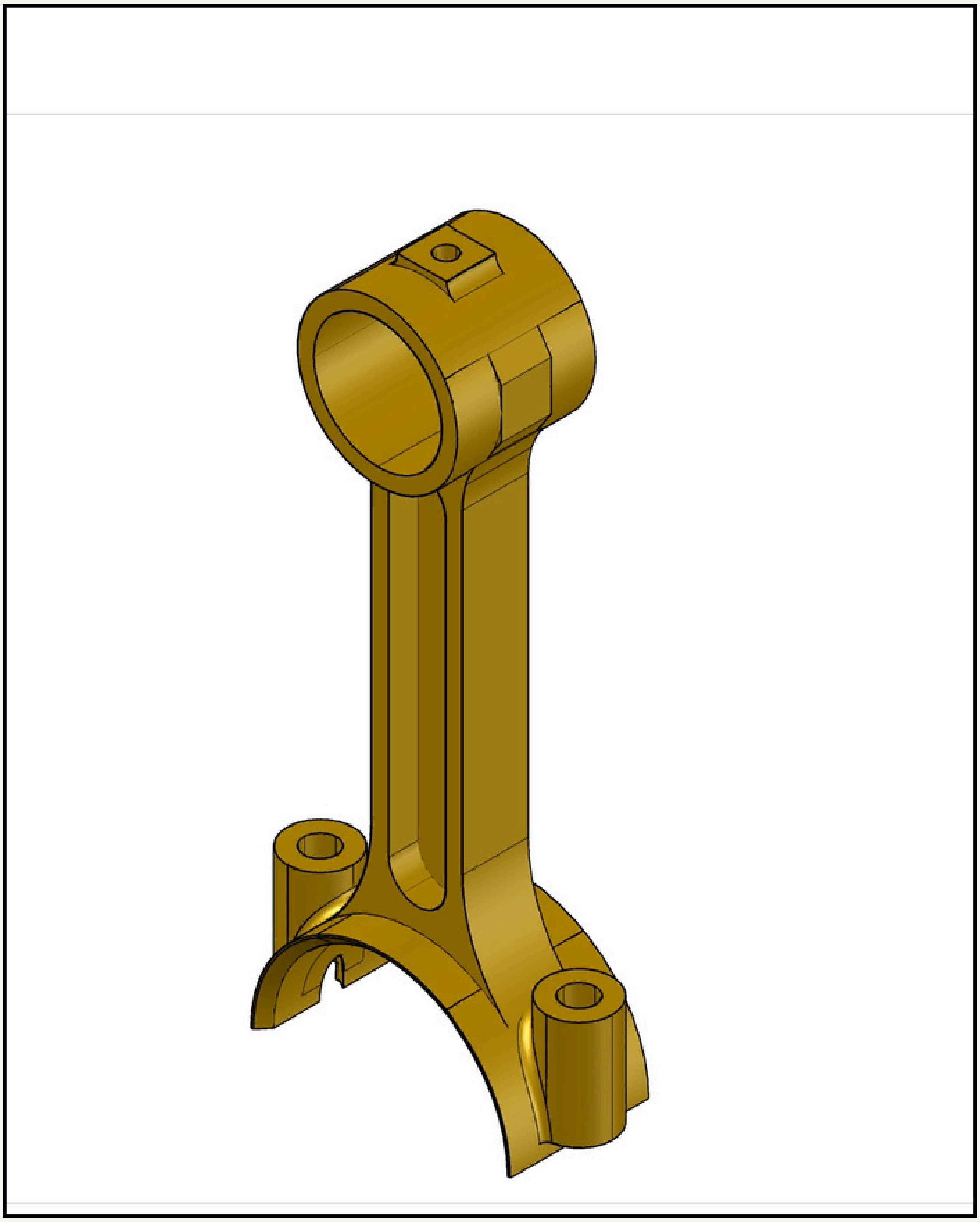

Figure 1: 3D Isometric View of Connecting Rod

2D Orthographic Views

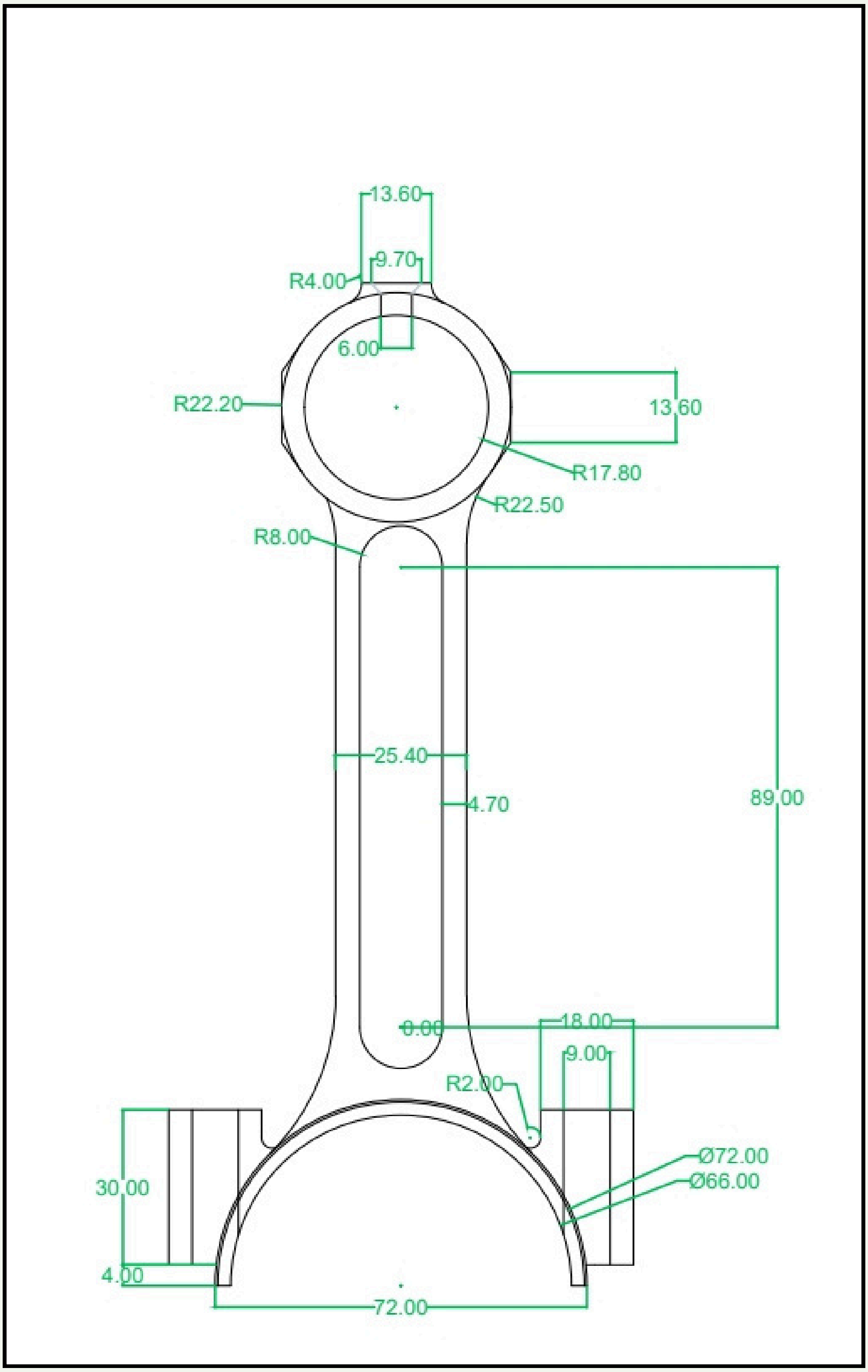

Figure 2: Front View

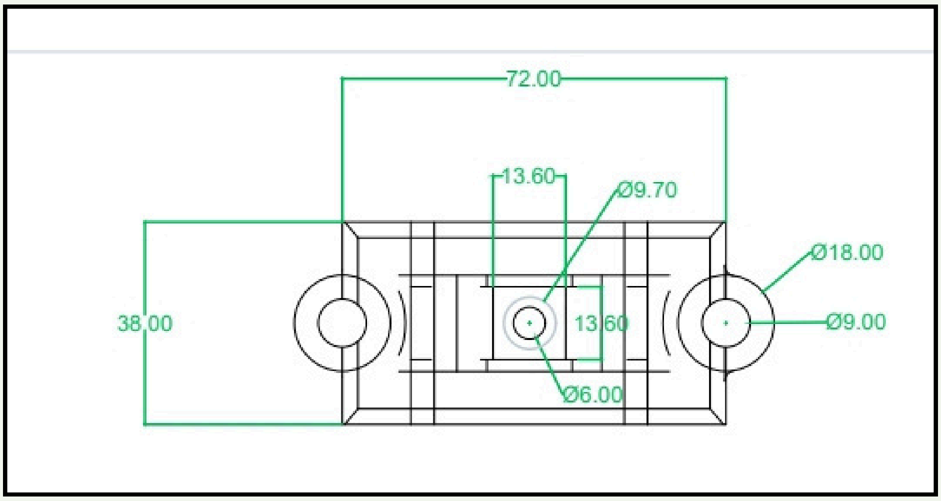

Figure 3: Top View

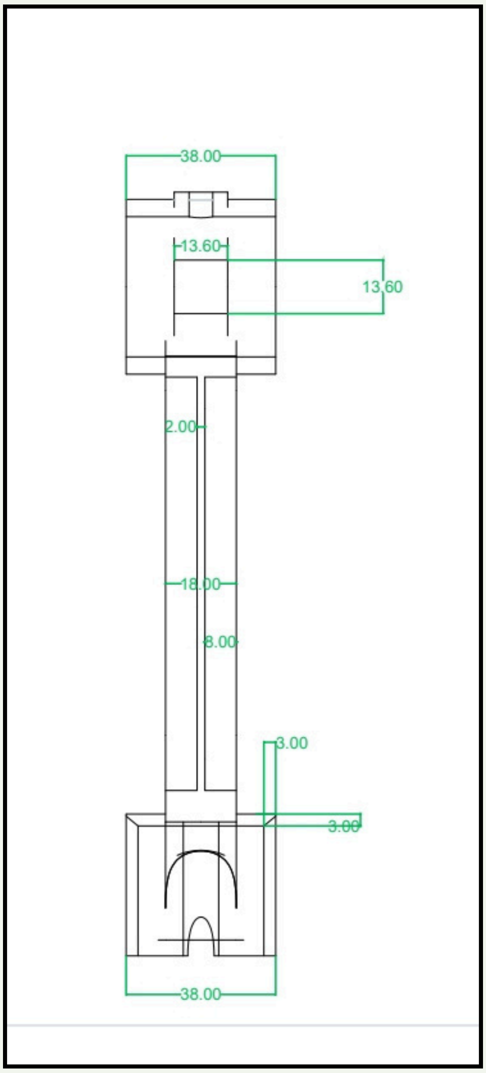

Figure 4: Side View

AutoCAD Commands Used

Command	Purpose
LINE	Create basic sketches
CIRCLE	Draw circular sections
ARC	Create curved geometry
OFFSET	Generate parallel profiles
TRIM	Remove unwanted portions
FILLET	Round edges
EXTRUDE	Convert 2D profile into 3D solid
PRESSPULL	Modify solid geometry
UNION	Combine solids
SUBTRACT	Remove material
FILLETEDGE	Create rounded edges
MOVE	Reposition objects
ROTATE	Rotate components
MIRROR	Create symmetrical geometry
VIEWBASE	Generate 2D views

Applications of Connecting Rod

- Automobile engines
- Motorcycle engines
- Diesel engines
- Air compressors
- Pumps
- Industrial machinery

Design Considerations

- Strength
- Weight reduction
- Fatigue life
- Manufacturability
- Cost efficiency

Result

A complete 3D model of a connecting rod was successfully created in AutoCAD. Orthographic views were generated and the model accurately represents the geometry of a practical connecting rod used in engines.